

Solving for Angle Measures on Parallel Lines

Answer Key

High-school geometry

Quick Answers

- | | |
|--|--|
| 1. 20 | degrees |
| 2. 30 | 7. (a) $x = 15$, (b) the measure of angle 2 = 100 |
| 3. (a) $x = 15$, (b) the measure of angle 3 = 63 | degrees |
| degrees | 8. (a) $x = 28$, (b) the measure of angle 5 = 116 |
| 4. 133 degrees | degrees |
| 5. (a) $x = 17$, (b) the measure of angle 4 = 63 | 9. 62 degrees |
| degrees | 10. (b) $x = 17$, (c) the measure of angle 8 = 92 |
| 6. (a) $x = 20$, (b) the measure of angle 6 = 115 | degrees, — |
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What is verified

9 of 10 problems fully machine-checked against SymPy.

— marks an answer only you can judge — problem 10. The worked solution states what a correct response must contain.

Curriculum

Level: High-school geometry

HSG-CO.C – *problems 1, 2, 3, 4, 5, 6, 7, 8, 9, 10*

Difficulty 1–5 of 5 across 17 tagged checks.

1. Corresponding angles $(2x + 14)^\circ$ and $(3x - 6)^\circ$, $j \parallel k$. Find x .

Solution: corresponding angles on parallel lines are congruent, so set the measures equal.

$$2x + 14 = 3x - 6$$

$$20 = x$$

Check: both angles measure $2(20) + 14 = 54$ and $3(20) - 6 = 54$.

$x = 20$

2. Same-side interior angles $(3x)^\circ$ and $(2x + 30)^\circ$, $j \parallel k$. Find x .

Solution: same-side interior angles are supplementary — add, do not equate.

$$3x + (2x + 30) = 180$$

$$5x + 30 = 180$$

$$5x = 150$$

Check: $3(30) = 90$ and $2(30) + 30 = 90$, and $90 + 90 = 180$.

$$x = 30$$

3. Alternate interior: $m\angle 3 = (5x - 12)^\circ$, $m\angle 6 = (3x + 18)^\circ$. (a) x . (b) $m\angle 3$.

(a) Alternate interior angles on parallel lines are congruent:

$$5x - 12 = 3x + 18$$

$$2x = 30$$

So $x = 15$.

$$x = 15$$

(b) $m\angle 3 = 5(15) - 12 = 75 - 12$. Check: $m\angle 6 = 3(15) + 18 = 63$ too.

$$63^\circ$$

4. $m\angle 1 = 47^\circ$, $j \parallel k$. Find $m\angle 7$.

Solution, two links: $\angle 1$ and $\angle 5$ are corresponding angles, so $m\angle 5 = 47$. Then $\angle 5$ and $\angle 7$ are a linear pair along the transversal at the lower vertex, so their measures add to 180: $m\angle 7 = 180 - 47$.

$$133^\circ$$

5. Same-side interior: $m\angle 4 = (4x - 5)^\circ$, $m\angle 6 = (6x + 15)^\circ$. (a) x . (b) $m\angle 4$.

(a)

$$(4x - 5) + (6x + 15) = 180$$

$$10x + 10 = 180$$

$$10x = 170$$

So $x = 17$.

$$x = 17$$

(b) $m\angle 4 = 4(17) - 5 = 68 - 5$. Check: $m\angle 6 = 6(17) + 15 = 117$, and $68 + 117 = 180$.

$$63^\circ$$

6. $m\angle 4 = (3x + 5)^\circ$, $m\angle 5 = (x + 45)^\circ$, alternate interior. (a) x . (b) $m\angle 6$.

(a)

$$3x + 5 = x + 45$$

$$2x = 40$$

So $x = 20$.

$$x = 20$$

(b) $m\angle 4 = 3(20) + 5 = 65$. Now $\angle 4$ and $\angle 6$ are same-side interior angles, so they are supplementary: $m\angle 6 = 180 - 65$. Check: $m\angle 5 = 20 + 45 = 65$, which agrees with $\angle 4$.

$$115^\circ$$

7. Alternate exterior: $m\angle 7 = (8x - 20)^\circ$, $m\angle 2 = (5x + 25)^\circ$. (a) x . (b) $m\angle 2$.

(a) Alternate exterior angles on parallel lines are congruent:

$$8x - 20 = 5x + 25$$

$$3x = 45$$

So $x = 15$.

(b) $m\angle 2 = 5(15) + 25 = 75 + 25$. Check: $m\angle 7 = 8(15) - 20 = 100$ too.

$$x = 15$$

$$100^\circ$$

8. Same-side interior: $m\angle 3 = (2x + 8)^\circ$, $m\angle 5 = (4x + 4)^\circ$. (a) x . (b) $m\angle 5$.
(a)

$$(2x + 8) + (4x + 4) = 180$$

$$6x + 12 = 180$$

$$6x = 168$$

So $x = 28$.

(b) $m\angle 5 = 4(28) + 4 = 112 + 4$. Check: $m\angle 3 = 2(28) + 8 = 64$, and $64 + 116 = 180$.

$$x = 28$$

$$116^\circ$$

9. Three parallel lines p , q , r cut by t ; the marked angle at p is 118° . Find $m\angle W$ at r .

Solution: corresponding angles carry the 118° down from p to q , and from q down to r — with three parallel lines the same transversal makes the same angle at each. The marked angle W at r sits between q and r on the other side of t from the carried angle, so the two are same-side interior and therefore supplementary: $m\angle W = 180 - 118$.

$$62^\circ$$

10. $m\angle 3 = (5x + 3)^\circ$, $m\angle 8 = (7x - 27)^\circ$. (a) Name the chain. (b) x . (c) $m\angle 8$.

(a) Two links. First, $\angle 3$ and $\angle 7$ are **corresponding angles** — both lie below their own line and on the same side of t — so with $j \parallel k$ they are congruent. Second, $\angle 7$ and $\angle 8$ are a **linear pair** along line k , so their measures add to 180. Putting the links together, $m\angle 3 = m\angle 7$ and $m\angle 7 + m\angle 8 = 180$, so $m\angle 3 + m\angle 8 = 180$: the two are *supplementary*, which is why the expressions must be added and not set equal.

(b)

$$(5x + 3) + (7x - 27) = 180$$

$$12x - 24 = 180$$

$$12x = 204$$

So $x = 17$.

(c) $m\angle 8 = 7(17) - 27 = 119 - 27$. Check: $m\angle 3 = 5(17) + 3 = 88$, and $88 + 92 = 180$.

$$x = 17$$

$$92^\circ$$

Grading (a): both link names, and the conclusion that a congruent link followed by a supplementary link leaves the endpoints supplementary.